

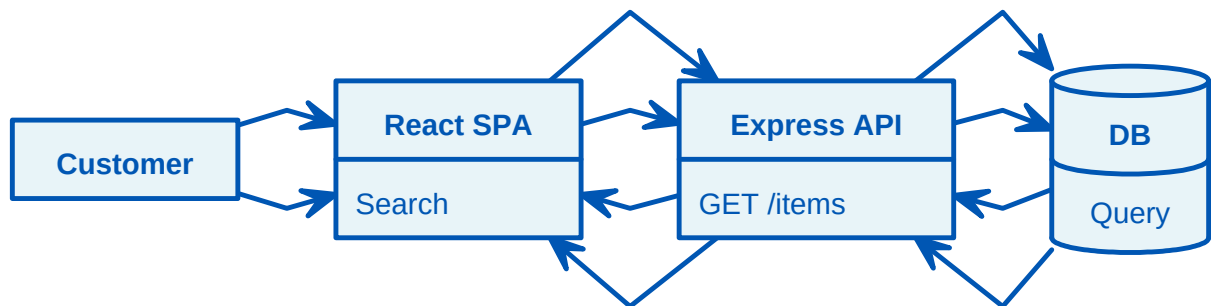
Data Flow Diagrams and User Stories

Date: 11 June 2026

Project Name: E-commerce Application

Data Flow Description

Sequence Diagram



- **Level 0 DFD (Context Diagram):** The Customer interacts with the "E-commerce System" by providing login details, search queries, and payment info. The system outputs product catalogs, order confirmations, and invoices. The Admin provides product updates and receives sales metrics.

- **Level 1 DFD:**

1. **Authentication Process:** Customer credentials -> Auth Service -> JWT Token.

2. **Shopping Process:** Customer Search -> Product DB -> Cart -> Checkout Service.
3. **Payment Process:** Checkout Service -> Payment Gateway (Payment Gateway) -> Order DB -> Confirmation to Customer.

User Stories

User Type	Epic	USN	User Story / Task	Acceptance Criteria	Priority	Release
Customer	Registration	USN-1	As a user, I can register by entering my name, email, and password.	I can access my account/dashboard upon successful registration.	High	Sprint-1
Customer	Login	USN-2	As a user, I can log in using my email and password.	System validates credentials and grants a JWT token session.	High	Sprint-1
Customer	Browsing	USN-3	As a user, I can search and filter products by category and price.	Search results reflect the applied filters accurately.	High	Sprint-1
Customer	Shopping Cart	USN-4	As a user, I can add, remove, or update quantities of items in my cart.	Cart total updates automatically; cart persists on reload.	High	Sprint-2
Customer	Checkout	USN-5	As a user, I can enter shipping details and pay via a payment gateway.	Order is saved, stock is deducted, and confirmation is shown.	High	Sprint-2
Customer	Reviews	USN-6	As a user, I can leave a review and rating for a product I bought.	Review appears on the product page and updates the average rating.	Medium	Sprint-3
Admin	Catalog	USN-7	As an admin, I can add, edit, or delete product listings and images.	Product changes reflect immediately on the storefront.	High	Sprint-2
Admin	Orders	USN-8	As an admin, I can view all orders and update their delivery status.	Status changes from "Processing" to "Delivered".	High	Sprint-3

Admin	Dashboard	USN-9	As an admin, I can view sales metrics and revenue charts.	Dashboard shows accurate monthly aggregation data.	Medium	Sprint-3
-------	-----------	-------	---	--	--------	----------